Positive Square Energy of Every Cyclomatic-Rank-Twelve Cactus

AI-generated manuscript; no human author is claimed

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Abstract

We prove that every connected cactus G of cyclomatic rank twelve on n vertices satisfies $s^+(G) > n$. The sharp cactus doubly nonnegative formula leaves exactly the cycle multisets $T^{11}Q$ and $T^{10}PP$, where $T = C_3$, $P = C_5$, and Q is an arbitrary cycle. The all-rank hostile and nonhostile one-cycle theorems close the first family for every parity of Q; the all-rank two-pentagon theorem closes the second. Only the exact frontier calculation is repeated here; the three rank-uniform structural proofs are cited rather than duplicated.

1 Sharp DNN bound and exact frontier

All graphs are finite, simple, and undirected. A cactus is connected and has only bridge or cycle blocks; its cyclomatic rank is the number of cycle blocks. For adjacency eigenvalues $\lambda_1, \dots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \qquad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \qquad \sigma(G) = s^+(G) - |V(G)|.$$

For an edge-containing graph define

$$\kappa(G) = \sup_{\substack{M \succeq 0,\ M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4\left(\sum_{uv \in E(G)} \sqrt{M_{uv}}\right)^2}{\mathbf{1}^T M \mathbf{1}}.$$

The sharp cactus DNN theorem [2, 1] states that if a cactus has b bridge blocks and cycle blocks $C_{\ell_1}, \dots, C_{\ell_r}$, then

$$s^-(G) \leq \kappa(G) = b + \sum_{i=1}^{r} \kappa(C_{\ell_i}), \qquad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even}, \\ \ell \sec^2\!\big(\pi/(2\ell)\big), & \ell \text{ odd}. \end{cases} \tag{1}$$

Set

$$\epsilon_\ell = \begin{cases} 0, & \ell \text{ even}, \\ \ell \tan^2\!\big(\pi/(2\ell)\big), & \ell \text{ odd}. \end{cases}$$

For rank twelve, $|E(G)| = n+11$ and block counting gives $b + \sum_i \ell_i = n+11$. Since $s^+(G) + s^-(G) = 2|E(G)|$, equation (1) yields the sharp-DNN surplus formula

$$\boxed{\sigma(G) \geq 11 - \sum_{i=1}^{12} \epsilon_{\ell_i}.} \tag{2}$$

Proposition 1.1 (Exact rank-twelve frontier)**.** *The right side of* (2) *is nonpositive exactly for*

$$\boxed{T^{11}Q} \quad (q \geq 3, \text{ including } Q = T), \qquad \boxed{T^{10}PP}.$$

Every other rank-twelve cycle multiset has positive surplus.

Proof. For real $x \geq 3$, $\epsilon_x = x\tan^2(\pi/(2x))$ decreases strictly: with $u = \pi/(2x)$, the derivative of $\tan^2 u/u$ has the sign of $2u - \sin u \cos u > 0$. Thus every nontriangle contributes at most

$$a := \epsilon_5 = 5 - 2\sqrt{5},$$

and the exact comparisons needed below are

$$0 < a < 1, \qquad 3a < 2, \qquad 2a > 1, \qquad a + \epsilon_7 < 1. \tag{3}$$

The first four inequalities after moving radicals to positive sides have squaring certificates $20 < 25$, $4 < 5$, $169 < 180$, and $80 < 81$. For the last, $\cos(\pi/7) > 2159/2401 > 7/8$ gives $\epsilon_7 < 7/15$, while $7/15 < 2\sqrt{5} - 4 = 1 - a$ follows from $4489 < 4500$.

Let t be the number of triangles. If $t \leq 9$, monotonicity and (3) give $\sum_i \epsilon_{\ell_i} \leq 9 + 3a < 11$. If $t = 10$, the two nontriangles contribute less than one when one is even, or when their odd lengths are not both five, since then their sum is at most $a + \epsilon_7 < 1$. The unique remaining pair is PP, and $10 + 2a > 11$. If $t \geq 11$, the multiset is $T^{11}Q$ and its excess sum is $11 + \epsilon_q \geq 11$. These cases are exhaustive. The exact note and fail-closed arithmetic verifier independently freeze this classification [3]. □

2 Closure of the frontier

Proposition 2.1. *Every connected cactus whose cycle blocks are eleven triangles and one cycle $Q = C_q$ satisfies $s^+(G) > |V(G)|$.*

Proof. If $q \equiv 1 \pmod 4$, then $q \geq 5$ and the all-rank triangle–hostile-cycle theorem applies [4]. If q is even or $q \equiv 3 \pmod 4$, the all-rank nonhostile one-cycle theorem applies [5]. These alternatives exhaust all integers $q \geq 3$, and the nonhostile theorem explicitly includes $q = 3$. Both theorems allow arbitrary shared cuts, bridge connectors, and finite tree attachments. □

Proposition 2.2. *Every connected cactus whose cycle blocks are ten triangles and two pentagons satisfies $s^+(G) > |V(G)|$.*

Proof. This is the all-rank two-pentagon theorem [6], specialized to $r = 10$. Its statement permits arbitrary cactus incidence, bridge lengths and branching, and arbitrary finite tree attachments. □

Theorem 2.3. *Every connected cyclomatic-rank-twelve cactus G on n vertices satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. Outside the two families in Proposition 1.1, the strict inequality follows immediately from (2). Proposition 2.1 closes $T^{11}Q$ for every parity of Q, and Proposition 2.2 closes $T^{10}PP$. The frontier classification is exhaustive, so the theorem follows. □

3 Reproduction, scope, and disclosure

From the public repository root, reproduce the exact frontier audit with Python 3.10 or newer:

```
python3 research/rank-twelve-cactus-dnn-residual-frontier-verifier.py
python3 -O research/rank-twelve-cactus-dnn-residual-frontier-verifier.py
```

Both runs report the certificate SHA-256 digest

`96d20340187cd0b2c01ca5d89d1f2c06cb0ecd321d035d73b53d94a706edd663`.

The SHA-256 digest of the cited verifier file itself is

`d94364685251afb9a1045085f6cd4b85774d37329bbf359a1c35dd738c3baae6`.

The verifier certifies only the sharp-DNN frontier; positivity on the two residual families comes from the cited all-rank manuscripts.

Scoped nonclaim. This paper proves only the stated result for finite simple connected cacti of cyclomatic rank twelve. It does not claim the positive-square-energy conjecture for all graphs, nor a theorem for arbitrary cactus rank or for non-cactus block intersections. The cited two-pentagon theorem concerns exactly triangles and two pentagons; it makes no claim for three pentagons.

AI disclosure

This manuscript was generated and assembled by an AI coding and mathematical reasoning system from the cited public-repository manuscripts, note, and exact verifier. No human authorship, human review, or human verification is claimed. The result should be assessed from the displayed reduction and the cited proof objects, not from asserted authority.

References

[1] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.

[2] Companion manuscript, *The Optimized Liu–Tang–Zhang Constant for Cactus Graphs*, 2026; `sharp-cactus-dnn/paper.tex`.

[3] *Exact rank-twelve cactus DNN residual frontier*, 30 July 2026; `research/rank-twelve-cactus-dnn-residual-frontier-2026-07-30.md` and `research/rank-twelve-cactus-dnn-residual-frontier-verifier.py`.

[4] Companion manuscript, *Cacti with Arbitrarily Many Triangles and One Hostile Cycle*, 2026; `all-rank-triangle-hostile-cacti/paper.tex`.

[5] *All-rank nonhostile one-cycle cactus theorem*, 30 July 2026; `research/all-rank-nonhostile-one-cycle-theorem-2026-07-30.md`.

[6] Companion manuscript, *Positive Square Energy for Cacti with Triangles and Two Pentagons*, 2026; `all-rank-triangle-two-pentagon-cacti/paper.tex`.